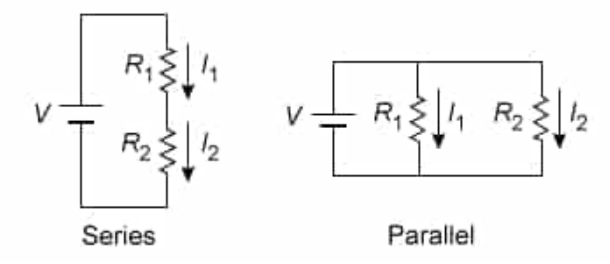
Two resistors, R_1 and R_2, are initially arranged in series. The same two resistors are then arranged in parallel, as shown.

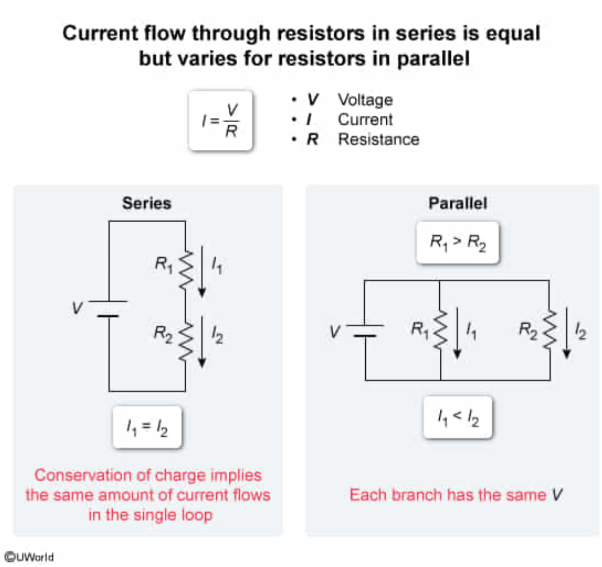


If $R_1 > R_2$, which of the following describes values the currents I_1 and I_2 through the resistors when they are arranged in series and then in parallel?

- A. $I_1 = I_2$ for the series configuration, and then $I_1 = I_2$ for the parallel configuration [4%]
- B. $I_1 = I_2$ for the series configuration, and then $I_1 < I_2$ for the parallel configuration [66%]
- C. $I_1 < I_2$ for the series configuration, and then $I_1 < I_2$ for the parallel configuration [12%]
- D. $I_1 > I_2$ for the series configuration, and then $I_1 = I_2$ for the parallel configuration [16%]

Correct
66% answered correctly

Explanation:



In an electric circuit, when circuit elements are connected end-to-end in a loop, they are in series, and **conservation of charge** dictates that the current *I* through all elements in series is equal:

$$I_1 = I_2 = \ldots = I_n$$

However, when circuit elements are arranged in parallel, the voltage ΔV is the same in each branch, but the current flow divides as it goes through the separate branches. Kirchhoff's junction rule follows from conservation of charge, implying that the sum of current entering the junction is equal to the sum of current leaving the junction:

$$\sum I_{\text{Entering junction}} = \sum I_{\text{Leaving junction}}$$

The amount of current in each branch is determined by the resistance R in each branch, as implied by Ohm's law, which states that I equals the ratio of V and R :

$$I = \frac{V}{R}$$

Thus, because I is inversely proportional to R, a lower resistance carries a higher current.

In this question, the resistors arranged in series must have the same current:

$$I_1 = I_2$$

However, the currents through each resistor can be different when they are arranged in parallel. Because $R_1 > R_2$ Ohm's law implies that:

$$I_1 < I_2$$

Therefore, the branch with the higher resistance R_1 has the lower current.

(Choices A, C, and D) The current is the same for both resistors when connected in series. However, when resistors are connected in parallel, the total current flow will divide into each branch; because $R_1 > R_2$, the current through R_1 is less than the current through R_2 , as implied by Ohm's law.

Educational objective:
When resistors are arranged in series, charge conservation implies the same amount of current flows through each resistor. When resistors are arranged in parallel, Ohm's law implies the branch with the greater resistance has the lowest current.

Time spent:0 sec
Subject:
Physics

QID:403916
System:
Electrostatics and Circuits